

CH # 11 Carbonyl compounds

Defination:-

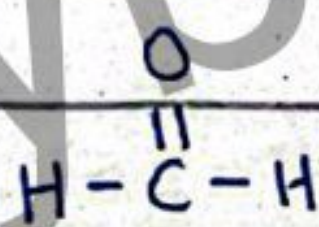
Organic compound containing carbonyl group (>C=O) are called carbonyl compounds.

- | | |
|------------------------|---------------------|
| (i) Aldehydes | (iv) Esters |
| (ii) Ketones | (v) Amides |
| (iii) Carboxylic Acids | (vi) Acid anhydride |

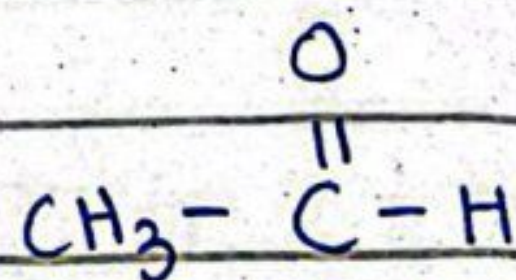
(i) Aldehydes:-

Organic compound containing carbonyl group directly attached with one hydrogen is called Aldehyde. $\text{C}_n\text{H}_{2n}\text{O}$

Example:-



Formaldehyde
or
Metanal

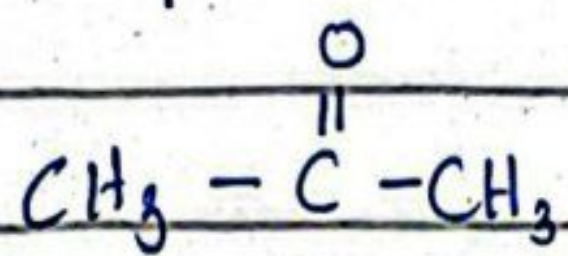


Acetaldehyde
or
Ethanal

(ii) Ketones:-

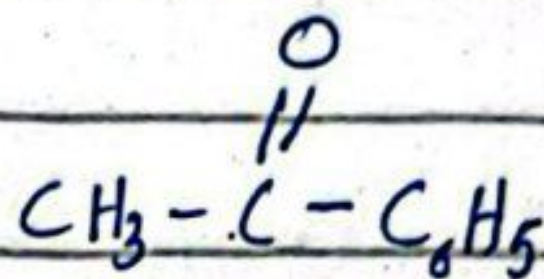
When carbonyl group is directly bonded with two alkyl or aryl group, ketones are formed. $\text{C}_n\text{H}_{2n}\text{O}$

Example:-



Acetone, Dimethyl ketone
or

Propanone



Methyl phenyl ketone
Acetophenone

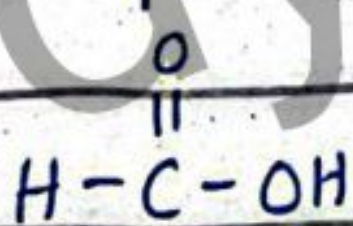
or

Phenylethanone

(iii) Carboxylic Acid:- $(-\overset{\text{O}}{\underset{\parallel}{\text{C}}}-\text{OH})$

when carbonyl group is directly bonded with hydroxyl group, compounds formed are called carboxylic acids. They are also called Alkanoic Acids. $(\text{C}_n\text{H}_{2n+1}\text{COOH})$ or $(\text{C}_n\text{H}_{2n}\text{O}_2)$

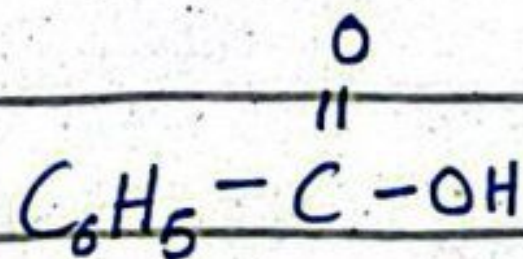
Examples:-



Formic Acid

or

Methanoic Acid

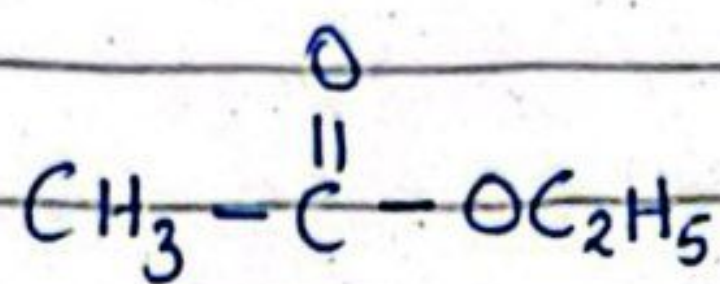


Benzoic Acid

(iv) Ester:-

when carbonyl group is directly bonded with alkoxy group compound formed is called Ester. $(\text{C}_n\text{H}_{2n+1}\text{COOR})$

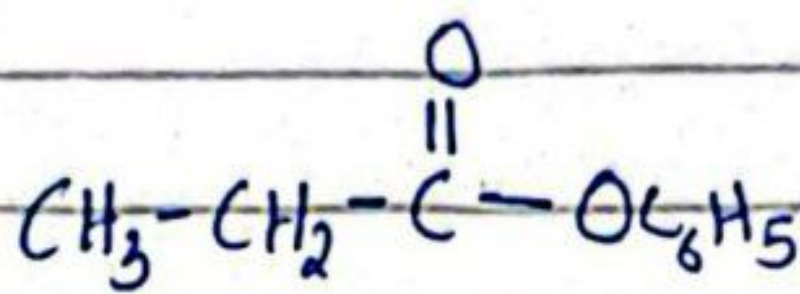
Example:-



Ethyl Acetate

or

Ethyl ethanoate



Phenyl propionate

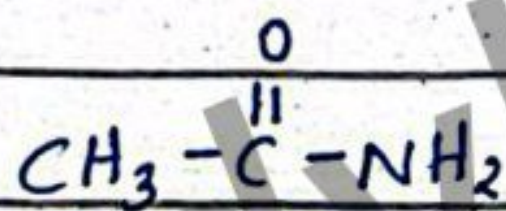
or

Phenyl propanoate

(v) Amides:- $(-\overset{\text{O}}{\parallel} \text{C}-\text{NH}_2)$

When carbonyl group is directly bonded with amino group Amides are formed. $\text{C}_n\text{H}_{2n+1}\text{COONH}_2$

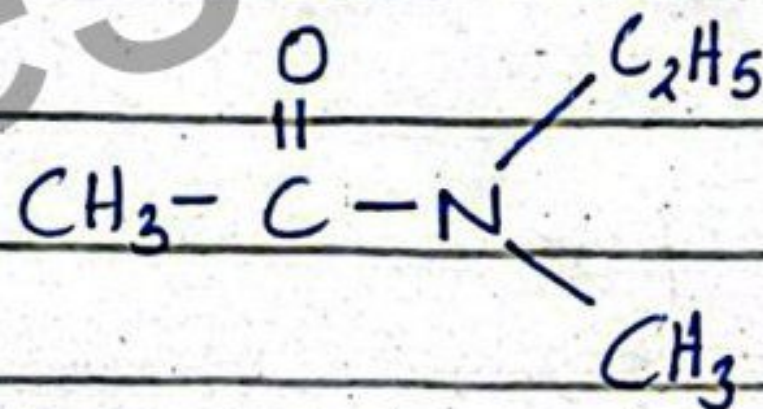
Example:-



Acetamide

or

Ethanamide



N-ethyl-N-methyl acetamide

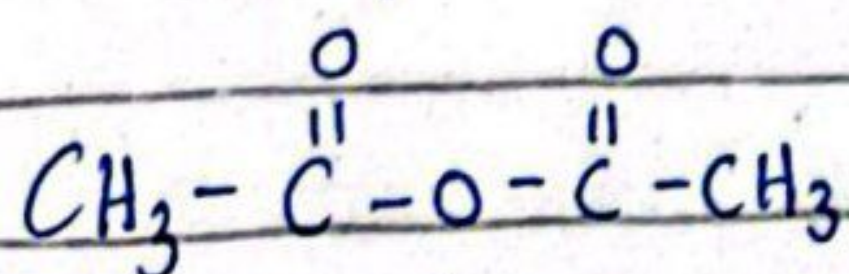
or

N-ethyl-N-methyl ethanamide

(vi) Acid anhydride:- $(-\overset{\text{O}}{\parallel} \text{C}-\text{O}-\overset{\text{O}}{\parallel} \text{C}-)$

When two molecules of carboxylic Acid are heated and water loss takes place Acid anhydride is formed. $(\text{C}_n\text{H}_{2n+1})_2\text{COO}$

Example:-



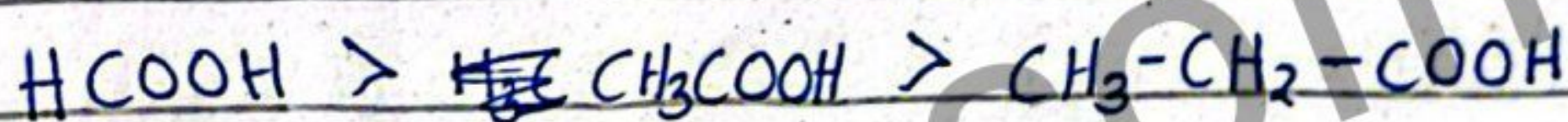
Acetic anhydride

or

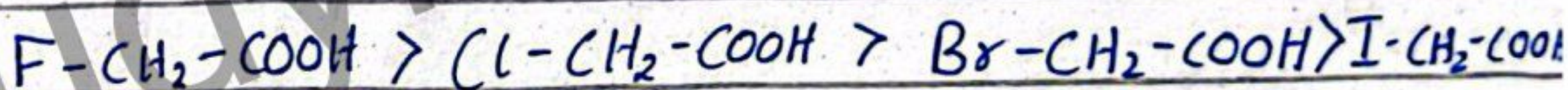
Ethanoic anhydride

Acidity:-

Acidity of $-\text{COOH} \propto \frac{1}{\text{No of Carbon atoms}}$



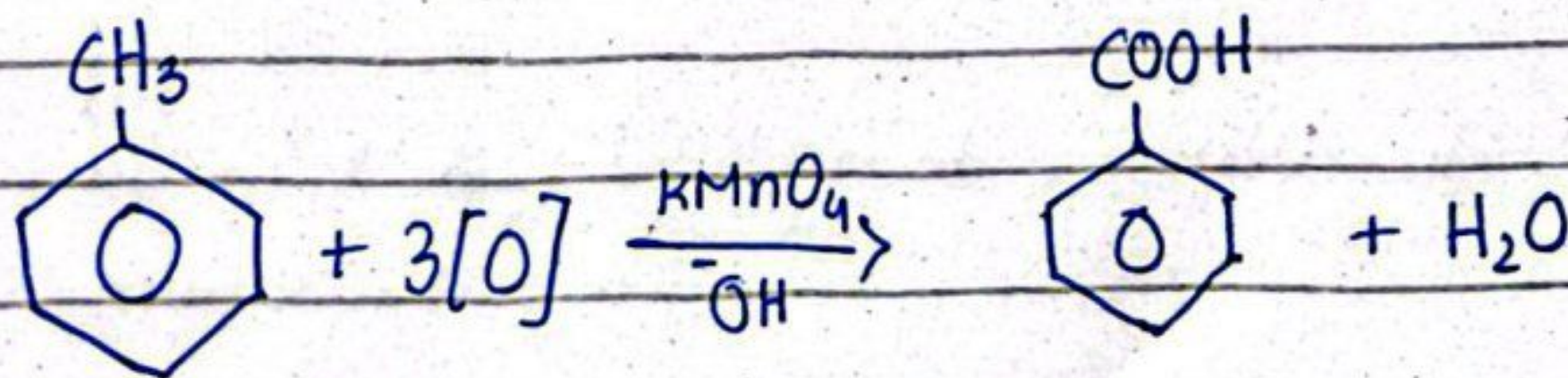
Acidity of $\text{X}-\text{COOH} \propto \text{No of X}^- \text{ atoms}$



Preparation of Benzoic Acid:-

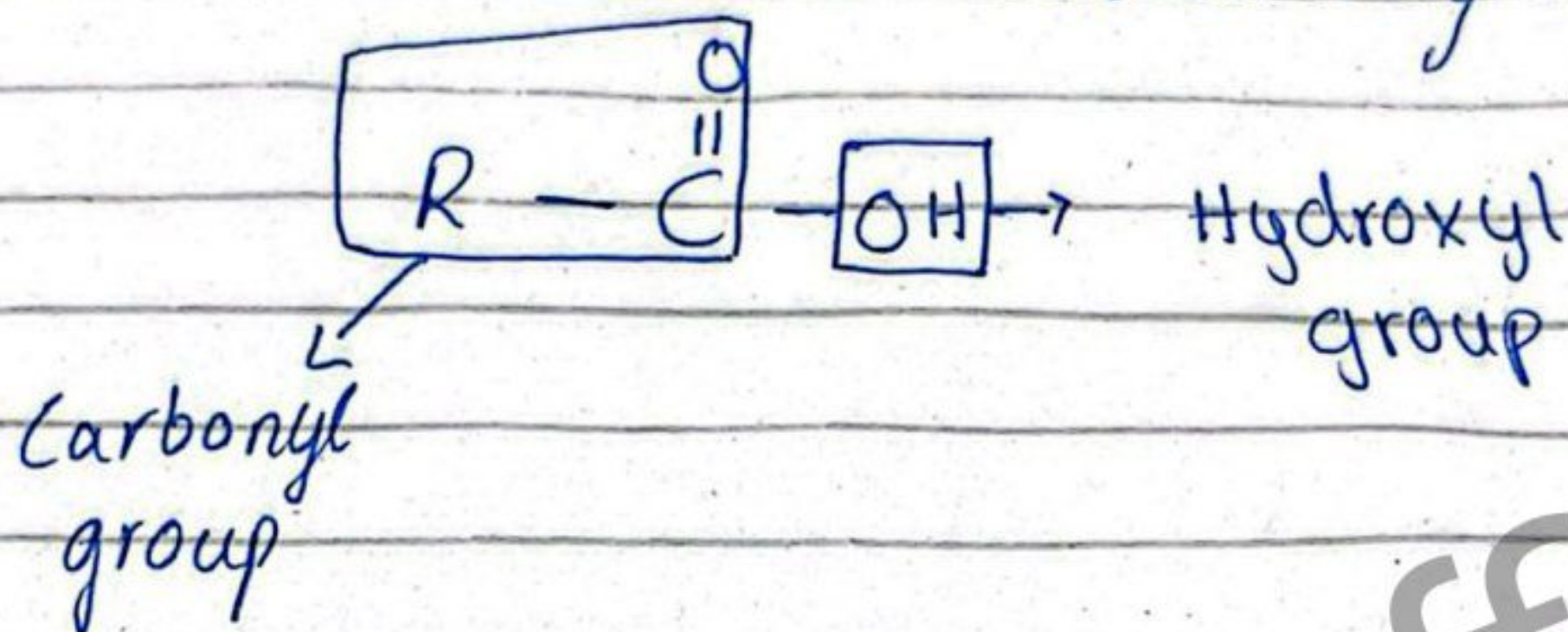
or

Oxidation of Alkyl Benzene:-



Reactivity of Carboxylic Acid:-

Carboxylic Acid are reactive due to presence of carbonyl group which increase the polarity of OH group due to electron withdrawing effect.



Reactions:-

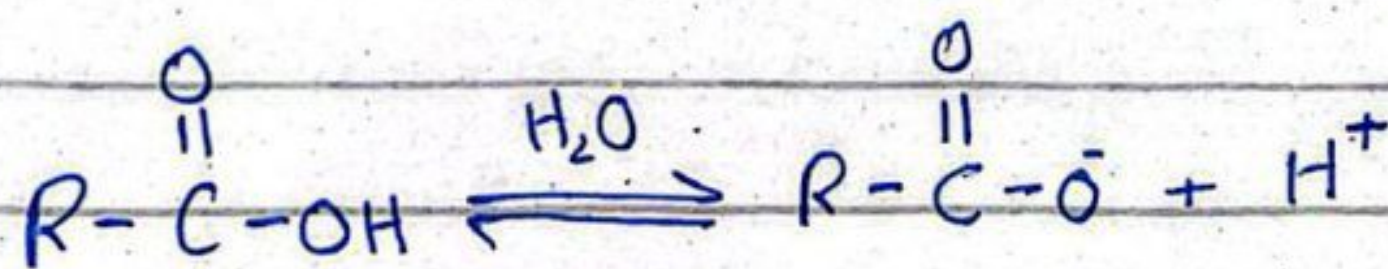
There are three types of reactions

1) Reaction in which H-atom of carboxylic group is involved.

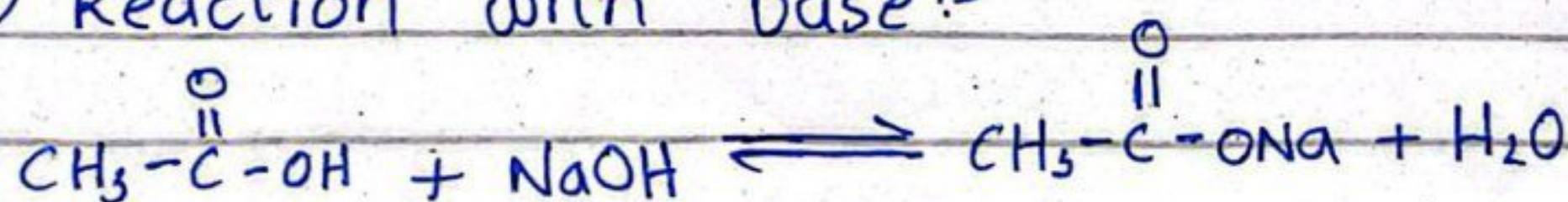
2) Reaction in which OH group of carboxylic group is involved

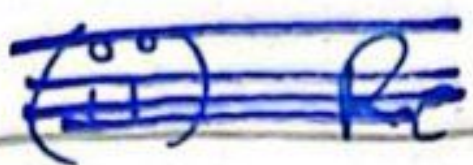
3) Reaction involving -COOH group.

1) Reaction with H-atom involved:-



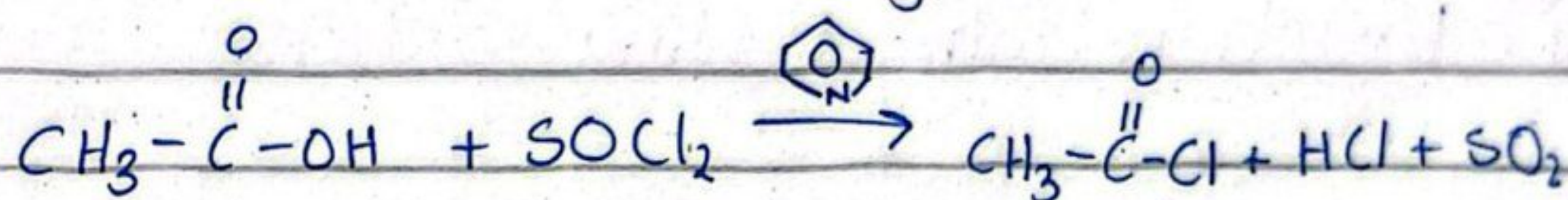
(i) Reaction with base:-



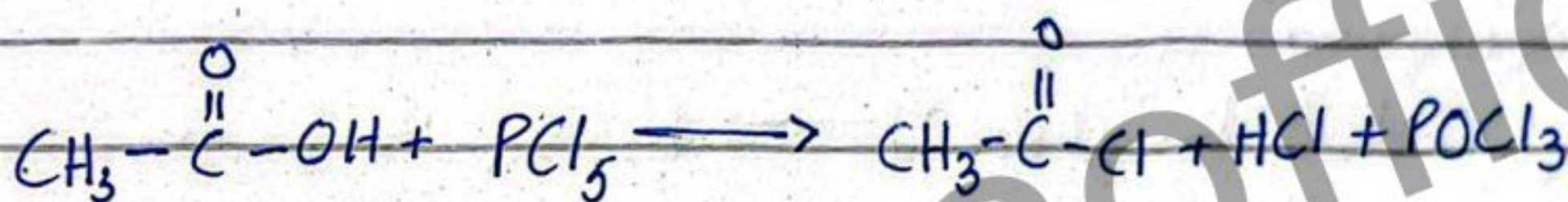
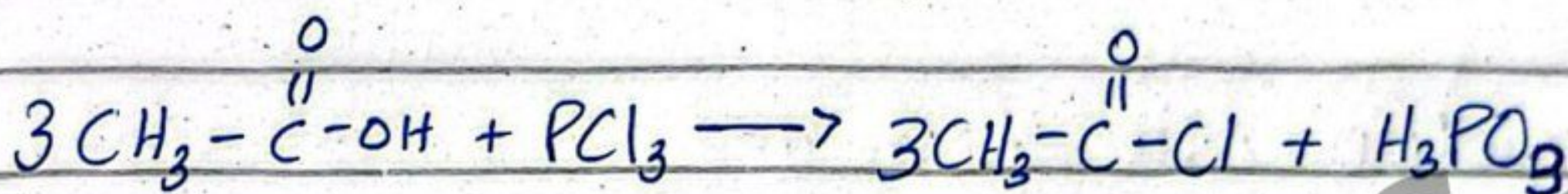


2) Reaction in which OH grp is involved:

(i) Reaction with thionylchloride

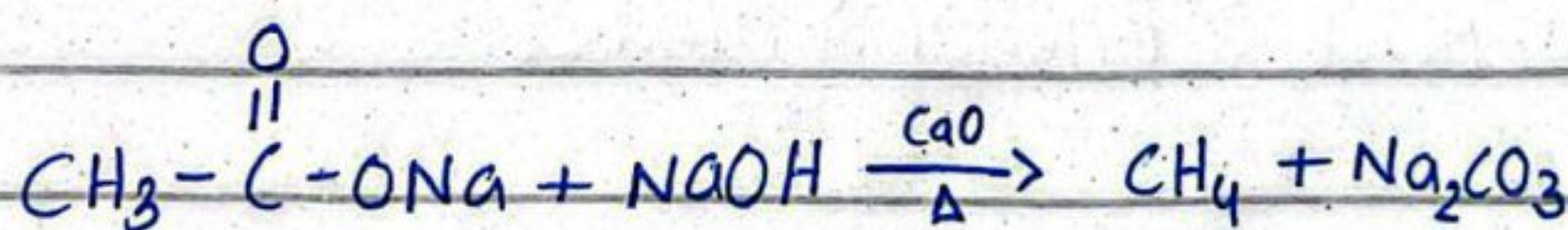
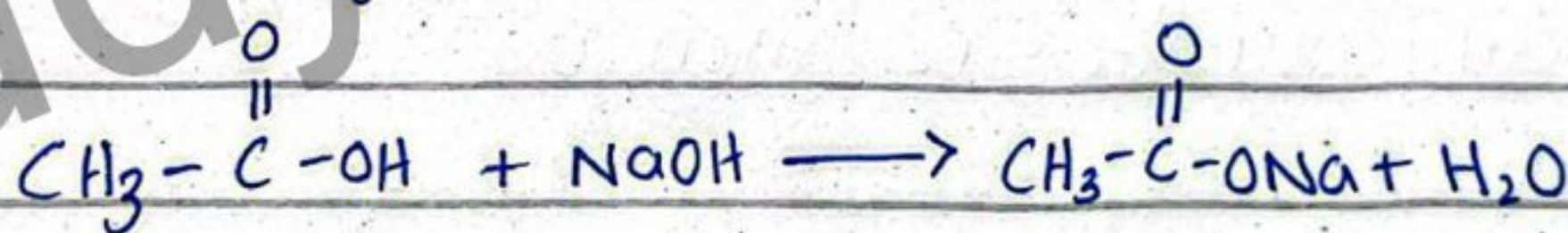


(ii) Reaction with phosphorus halide



3) Reaction involving $-\text{COOH}$ group:-

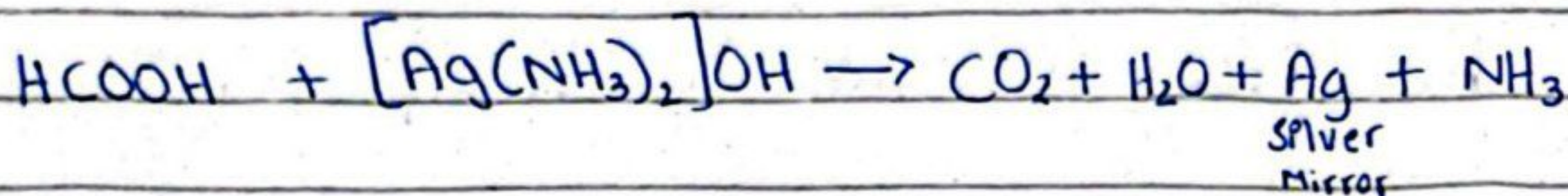
(i) DeCarboxylation



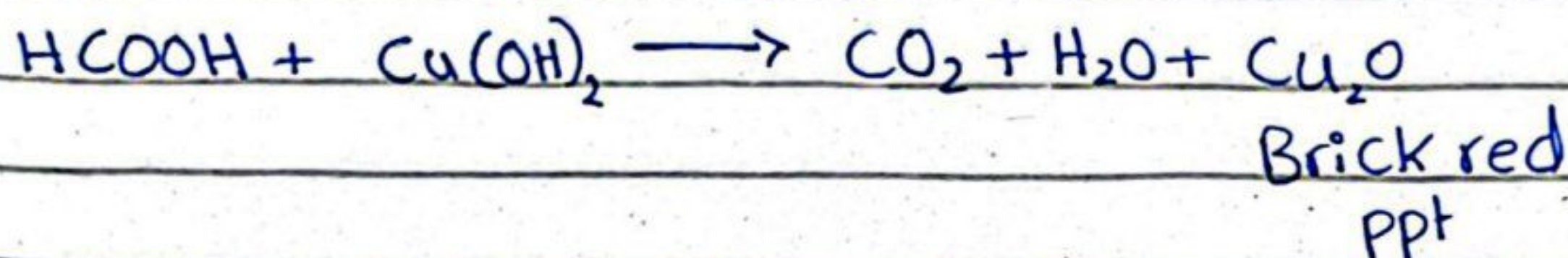
Oxidation Reactions:-

Some carboxylic Acid undergo oxidation reaction like formic and oxalic acid.

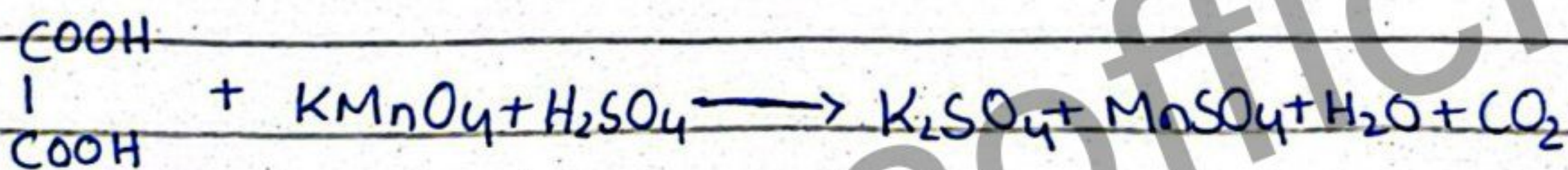
(i) Reaction with Tollen's reagent:- ($\text{AgNO}_3 + \text{NH}_4\text{OH}$)



(ii) Reaction with Fehling solution:- ($\text{CuSO}_4 + \text{NaOH} + \text{Tartaric Acid}$)



(iii) Reaction with KMnO_4 :-



Comparison of acidity of Phenol, Alcohol and Carboxylic Acid.

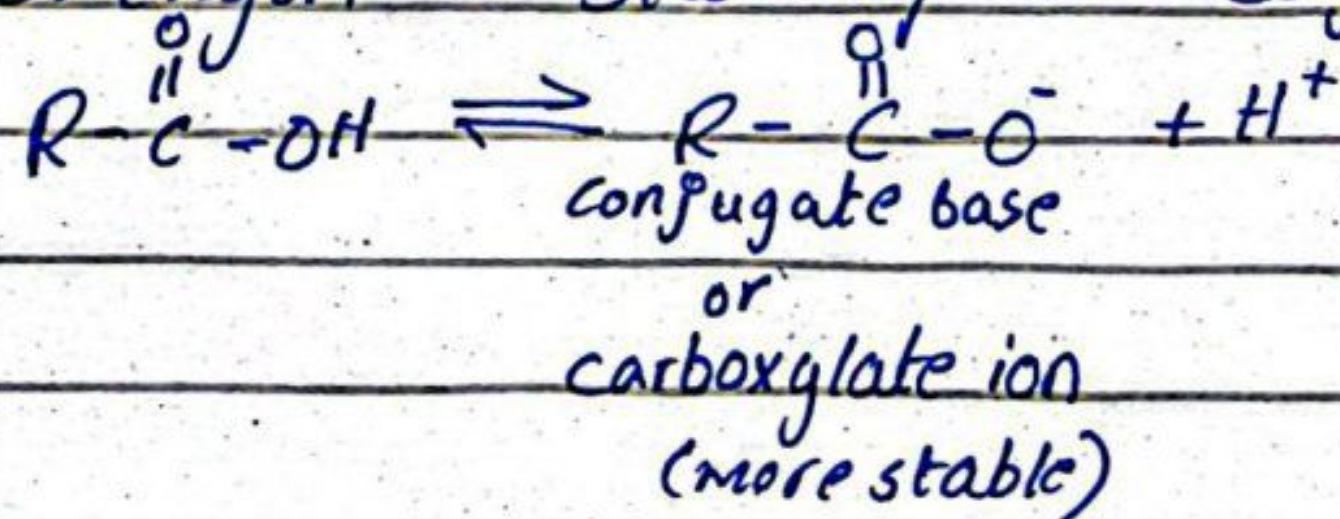
1) Acidity order is given by

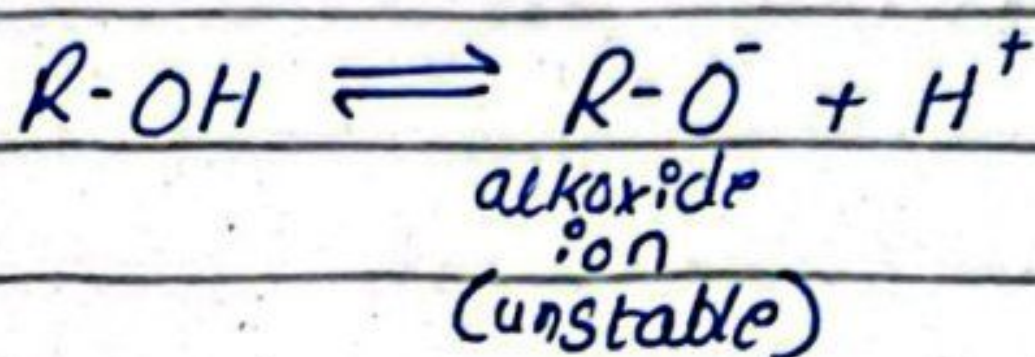
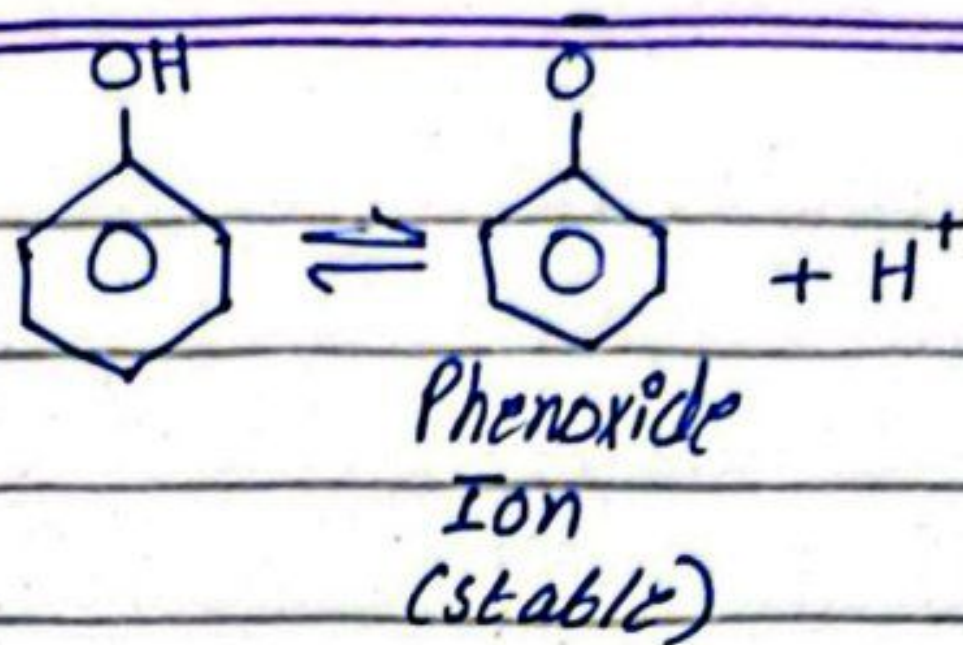
Carboxylic Acid > Phenol > Alcohol

2) $pK_a \propto \frac{1}{\text{Strength of Acid}}$

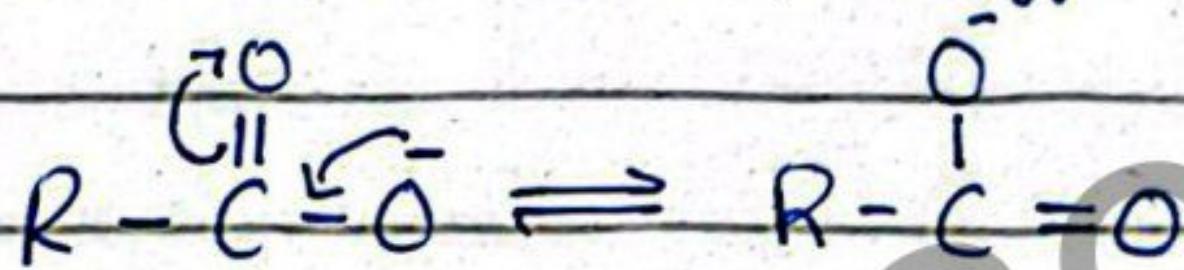
pK_a of carboxylic acid ≈ 5
 pK_a of phenol = 10
 pK_a of Alcohol = 16 to 19

3) Acidic strength $\propto$ stability of conjugate base

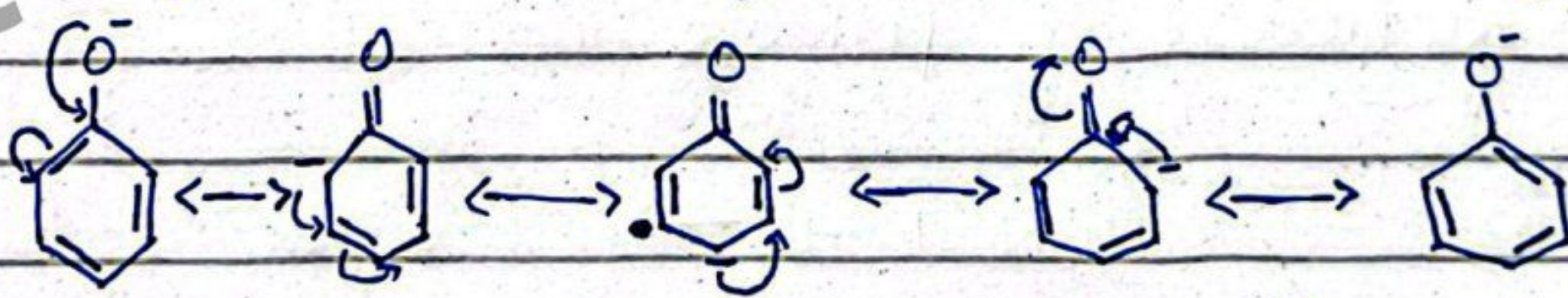




4) Carboxylic Acid is stronger than phenol because in carboxylic acid -ve charge delocalize from one oxygen atom to another

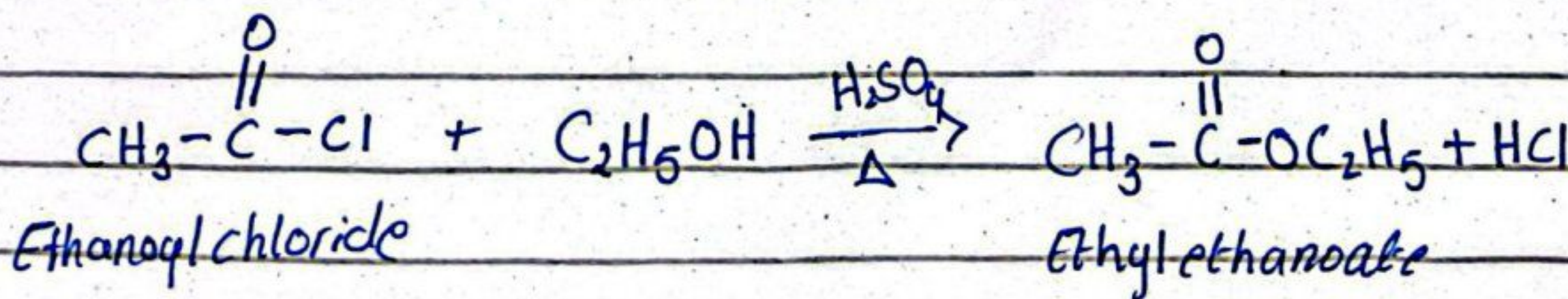


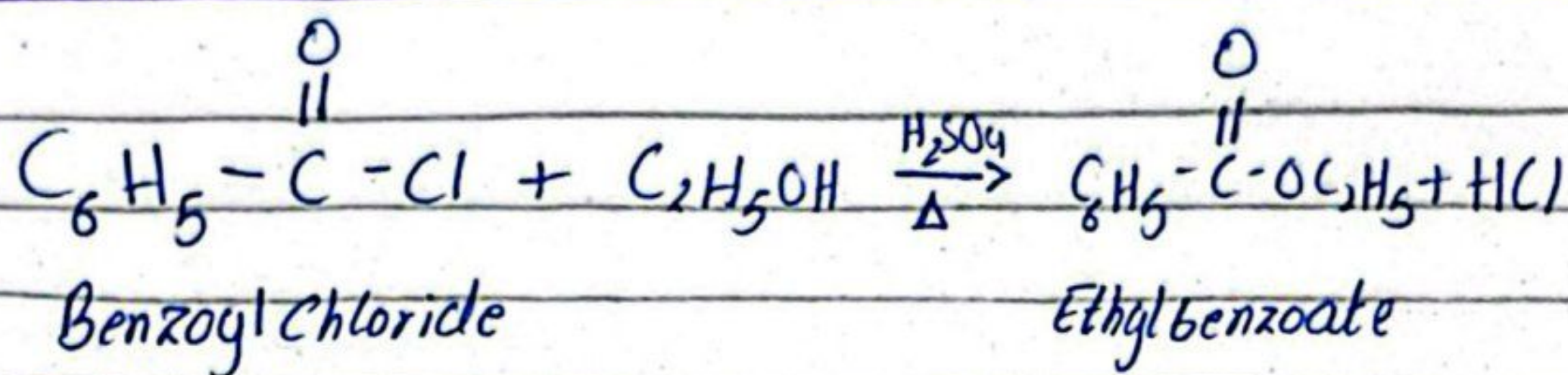
5) Phenol is stronger acid than alcohol because in phenol phenoxide ion is stable due to delocalization of -ve charge from oxygen to benzene ring.



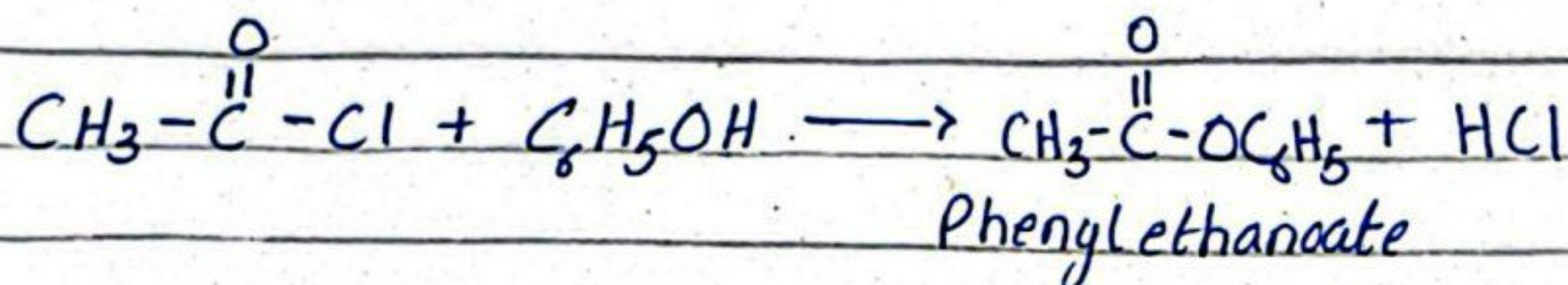
Reaction of Acylchloride:- $(\text{R}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Cl})$

1) Reaction with alcohol:-

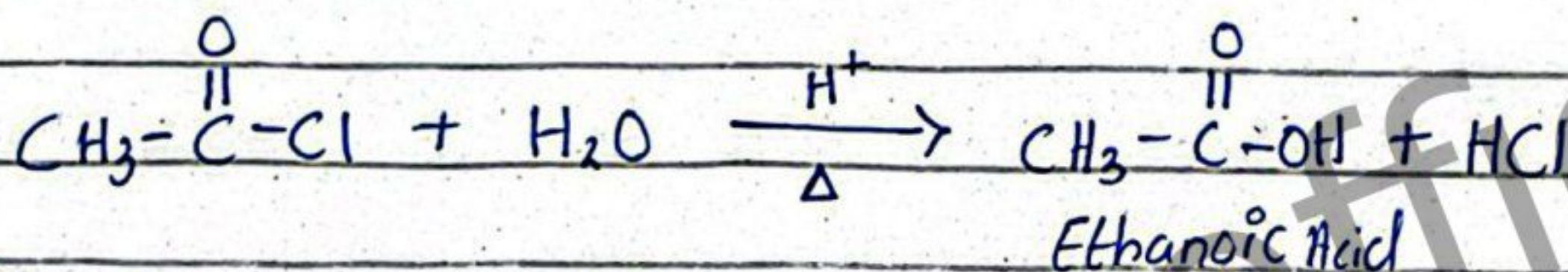




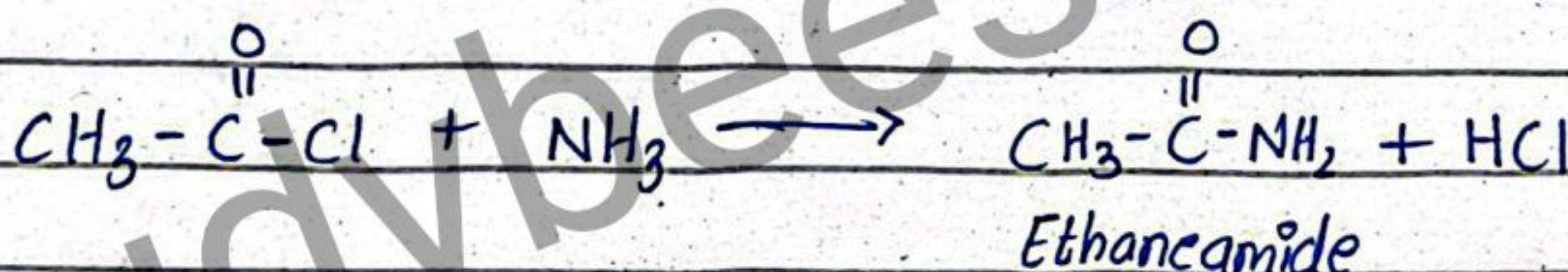
2) Reaction with Phenol:-



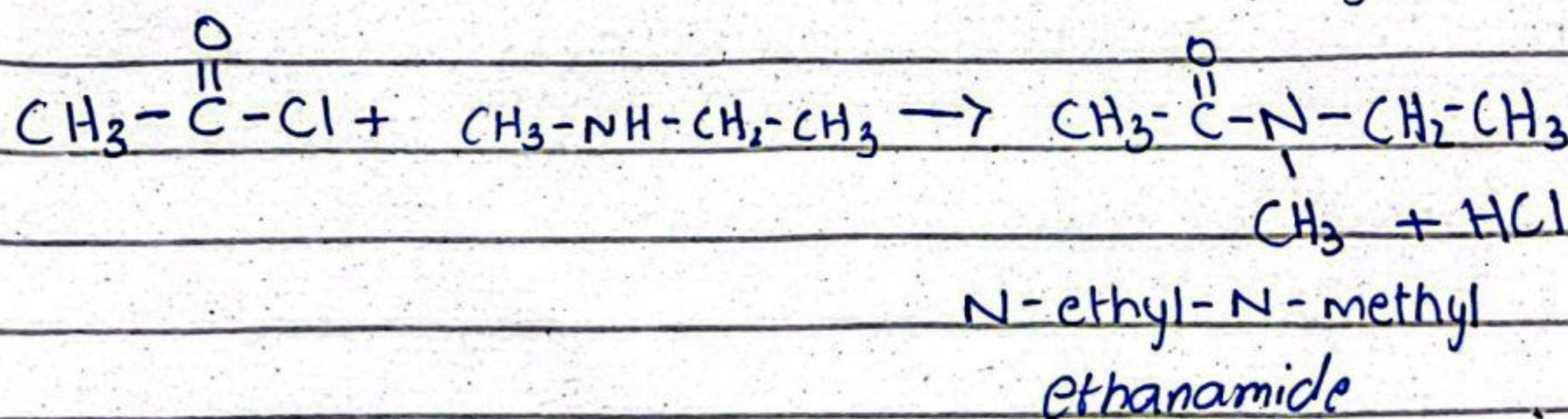
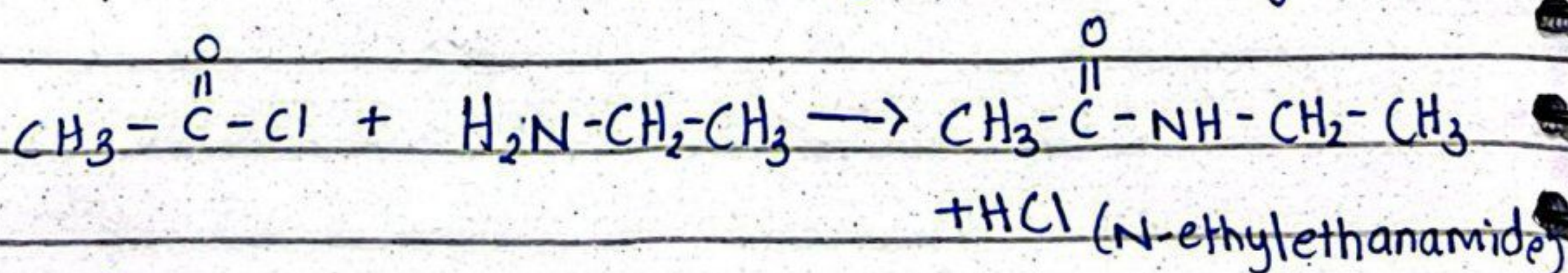
3) Reaction with water:-



4) Reaction with Ammonia:-

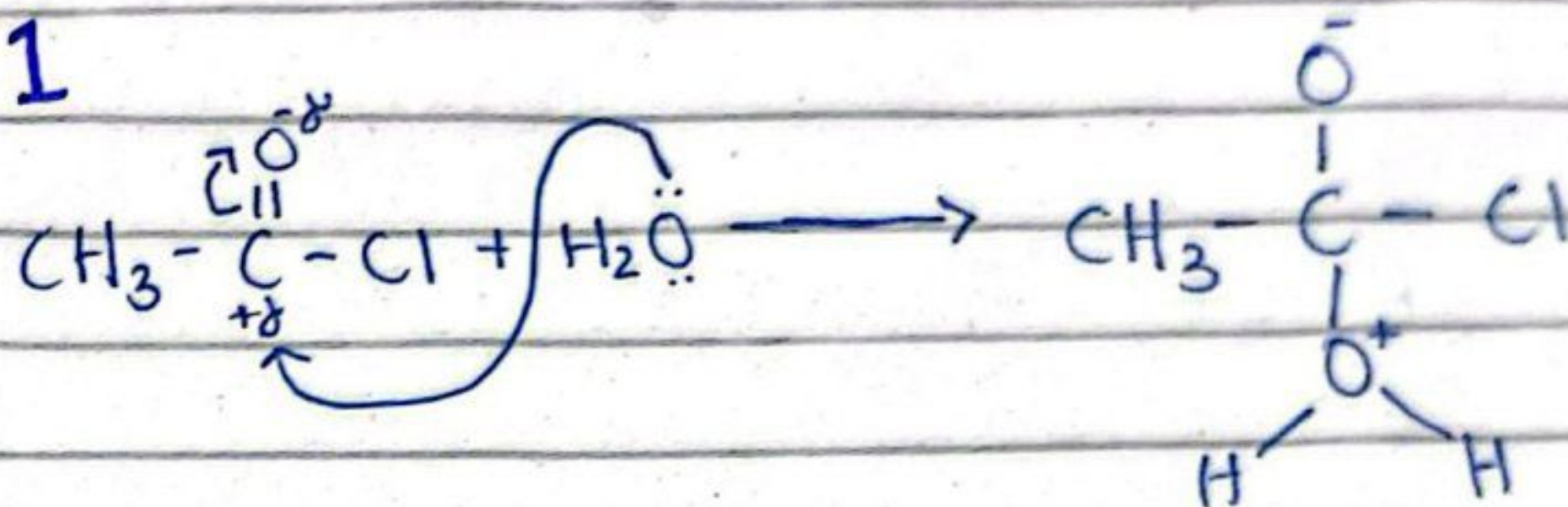


5) Reaction with primary and secondary amines:-

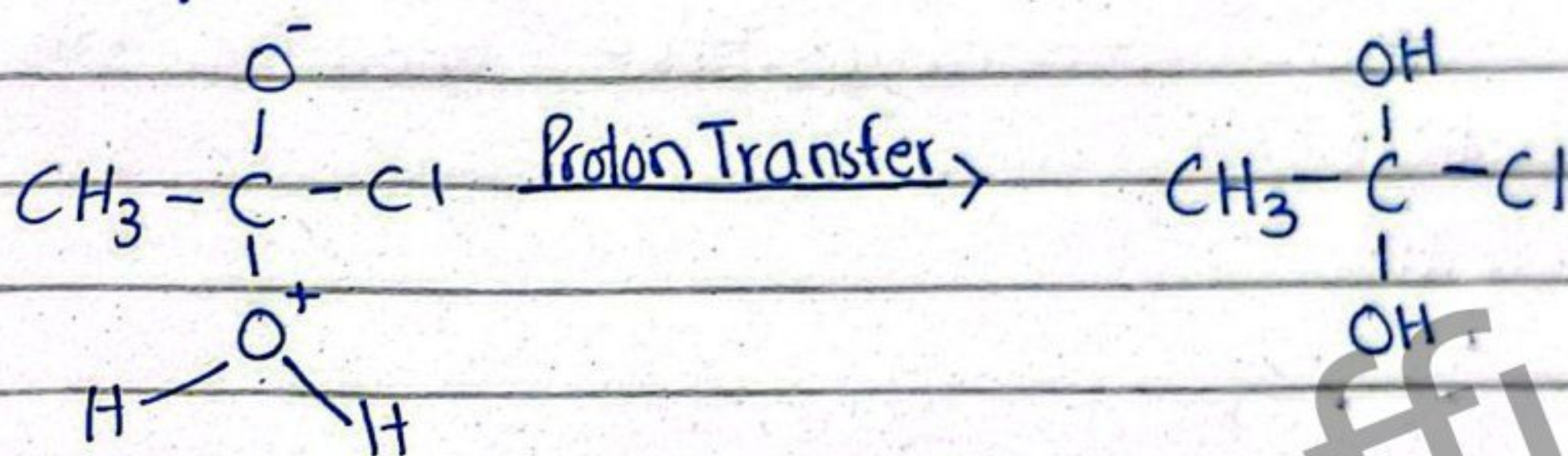


Mechanism of Acylchloride with Nucleophile

Step-1



Step-2



Step-3



Comparison of hydrolysis Acyl chloride, Alkyl chloride and aryl chloride:-

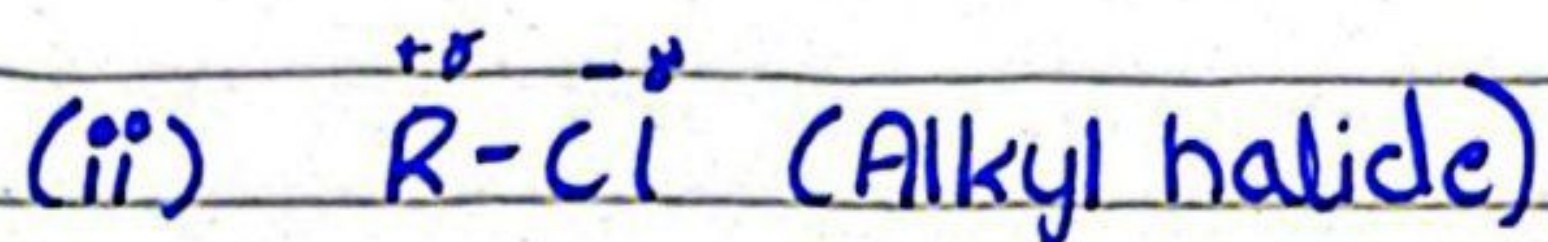
Ease of hydrolysis order:-

Acyl chloride > Alkyl halide > Aryl chloride

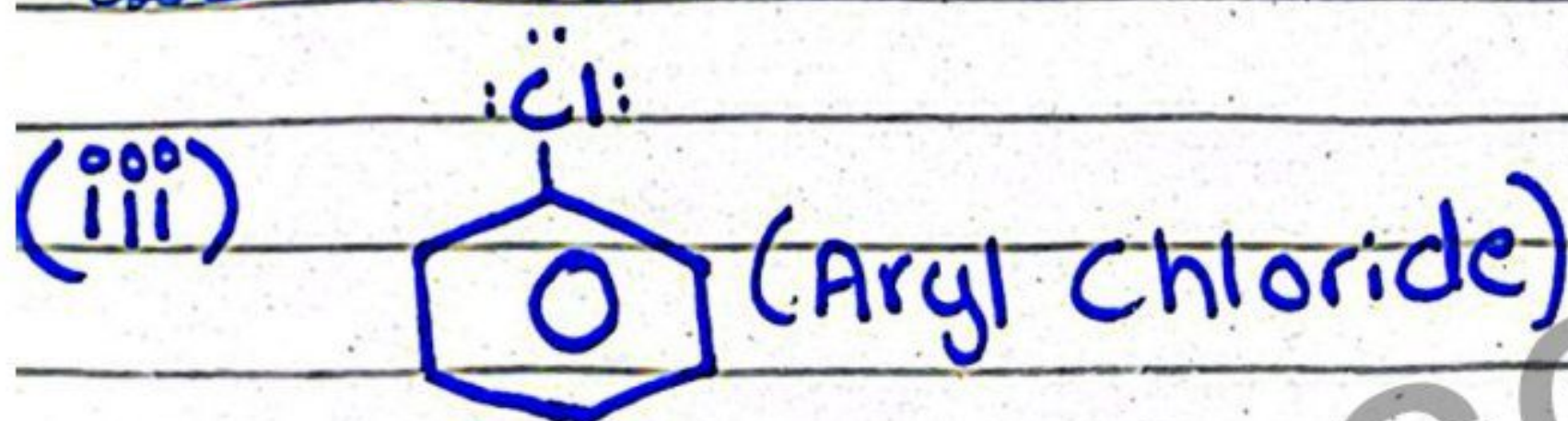
(i) $\text{R}-\overset{\overset{\text{O}}{\parallel}}{\text{C}}-\text{Cl}$ (Acyl chloride)

In Acyl chloride carbonyl carbon is bonded

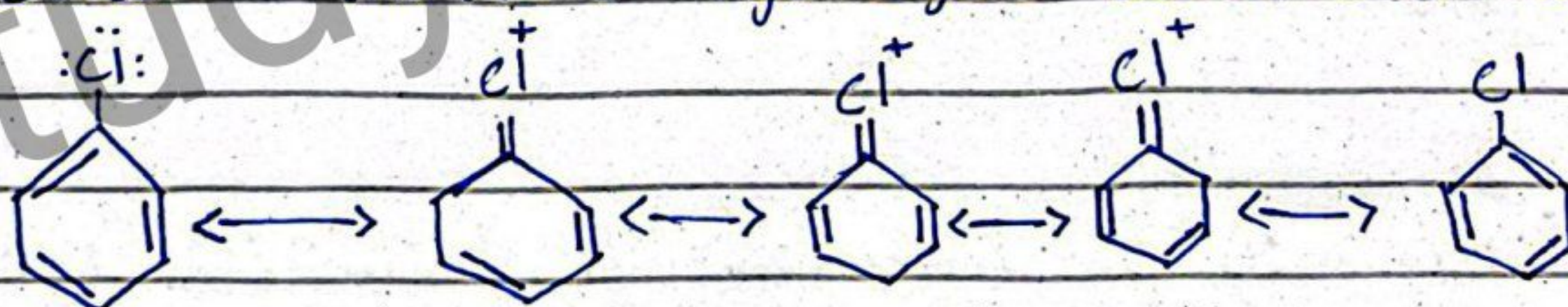
with two electronegative atoms oxygen and chlorine hence carbonyl carbon is more electropositive.



In alkyl halide carbon is bonded to one electronegative atom chlorine hence it is less electropositive than carbonyl carbon.



In aryl chloride lone pair of chlorine is delocalized and becomes more stable therefore hydrolysis becomes difficult.



KEY POINTS

- Carboxylic acids are the most acidic simple organic compounds ($pK_a \approx 5$).
- The electron-withdrawing group bonded to the carbon atom next to the carboxyl group ($COOH$) makes the acid stronger.
- The carboxyl group shows the chemistry of both the carbonyl ($\text{—}\overset{\text{O}}{\text{C}}\text{—}$) and the hydroxyl of an ($-OH$) group.
- The $-OH$ group of carboxylic acids can be replaced by X , OR , and NH_2 to form halides, esters, and amides respectively.
- Phosphorus(III)chloride reacts with carboxylic acids at room temperature producing acyl chloride and phosphorus acid.
- Methanoic acid is oxidized to carbon dioxide and water by Tollen's reagent with the formation of a silver mirror.
- Methanoic acid is also oxidized to carbon dioxide and water by Fehling's solution with the formation of red precipitates of copper(I)oxide.
- Phenoxide ion is stabilized due to resonance. Ethoxide ion on the other hand has no delocalization of the negative charge.
- Acyl halides react vigorously with alcohols at room temperature to produce esters.
- Acyl chloride reacts with phenol at room temperature to produce an ester.
- Acyl halides hydrolyse with water at room temperature to produce carboxylic acids and hydrochloric acid.
- Acyl halides react with ammonia at room temperature to produce amides.

EXERCISE

1. Multiple Choice Questions (MCQs)

- Which reagent would you use to convert toluene to benzoic acid?
 - ☒ a) $KMnO_4$ and dilute acid
 - b) Fehling's solution
 - c) PCl_5
 - d) $SOCl_2$
- What is the main product of the reaction between acetic acid and PCl_5 ?
 - a) Ethyl acetate
 - ☒ b) Acetyl chloride
 - c) Acetic anhydride
 - d) Acetone
- Which of the following is a strong oxidizing agent for the oxidation of methanoic acid?
 - a) PCl_5
 - b) $SOCl_2$
 - ☒ c) $KMnO_4$
 - d) PCl_3
- In terms of acidity, which of the following is the most acidic?
 - a) Methanol
 - b) Phenol
 - ☒ c) Formic acid
 - d) Benzoic acid
- What happens when benzoyl chloride reacts with water?
 - ☒ a) Benzoic acid and HCl are formed.
 - b) Benzene and water are formed.
 - c) Phenol and water are formed.
 - d) Benzoate and water are formed.

- vi. The mechanism by which acyl chlorides react with alcohols is:
- a) Addition
 - b) Elimination
 - c) Addition-elimination
 - ✓d) Substitution
- vii. Which of the following acyl chlorides is expected to hydrolyze most readily?
- a) Methyl chloride
 - ✓b) Acetyl chloride
 - c) Phenyl chloride
 - d) Benzyl chloride
- viii. When formic acid is oxidized, which of the following is formed?
- a) Water only
 - b) Carbon dioxide only
 - ✓c) Methanol only
 - ✓d) Water and carbon dioxide
- ix. The relative acidity of chlorinated carboxylic acids is primarily due to:
- a) Increased stability of the conjugate base.
 - b) Electron-donating effects of the chlorine atom.
 - ✓c) Lowering of the carboxylate pK_a .
 - d) Increased electron density around the carbonyl group.
- x. The addition-elimination reaction mechanism of acyl chlorides involves:
- ✓a) Nucleophilic attack on the carbonyl carbon.
 - b) Addition of HCl to the carbonyl group.
 - c) Electrophilic substitution on the benzene ring.
 - d) Radical addition to the carbonyl group.
- xi. Why are carboxylic acids more acidic than phenols and alcohols?
- a) Carboxylic acids have a lower molecular weight.
 - b) Carboxylic acids are better at forming hydrogen bonds.
 - ✓c) The carboxylate anion is resonance-stabilized.
 - d) Carboxylic acids have a higher boiling point.
- xii. How does chlorine substitution affect the acidity of carboxylic acids?
- a) It increases acidity by donating electrons.
 - b) It decreases acidity by withdrawing electrons.
 - ✓c) It increases acidity by withdrawing electrons.
 - d) It decreases acidity by donating electrons.
- xiii. Which of the following statements about the addition-elimination mechanism of acyl chlorides is correct?
- ✓a) The nucleophile attacks the carbonyl carbon, forming a tetrahedral intermediate.
 - b) The nucleophile attacks the chlorine atom directly.
 - c) The reaction involves a free radical intermediate.
 - d) The acyl chloride first forms a carbocation intermediate.

- xiv. What is the major difference in the oxidation products of ethanedioic acid and methanoic acid when treated with warm acidified KMnO_4 ?
- Ethanedioic acid produces carbon dioxide, while methanoic acid produces formaldehyde.
 - Ethanedioic acid produces carbon monoxide, while methanoic acid produces carbon dioxide and water.
 - ✓ c) Ethanedioic acid produces carbon dioxide, while methanoic acid produces carbon dioxide and water.
 - d) Ethanedioic acid produces acetic acid, while methanoic acid produces carbon dioxide.

2. Short Answer Questions

- Write the equation for the reaction of acetic acid with SOCl_2 .
- Explain why methanoic acid can be oxidized while other carboxylic acids cannot.
- What products are formed when ethanedioic acid is oxidized with warm acidified KMnO_4 ?
- Describe the mechanism of acyl chloride hydrolysis with water.
- What is the difference in reactivity between acyl chlorides and alkyl chlorides?
- Explain the influence of chlorine substitution on the acidity of carboxylic acids.
- What product results from the reaction of acyl chlorides with phenol?
- Methylbenzene can be oxidized to benzoic acid using hot, alkaline KMnO_4 and then dilute acid.
 - Describe the role of potassium permanganate in the oxidation of methylbenzene.
 - Explain why the reaction mixture needs to be acidified after oxidation.
- Carboxylic acids can be converted to acyl chlorides using PCl_3 , PCl_5 or SOCl_2 .
 - Write the balanced chemical equation for the reaction of acetic acid with PCl_5 .
 - Compare the by-products formed when using PCl_3 and SOCl_2 for this conversion.
- Methanoic acid can be oxidized by Fehling's solution, Tollen's reagent, acidified KMnO_4 , or acidified $\text{K}_2\text{Cr}_2\text{O}_7$.
 - Outline the changes observed when methanoic acid is treated with Tollen's reagent.
 - Write the balanced chemical equation for the oxidation of methanoic acid with acidified KMnO_4 .
- Carboxylic acids are generally more acidic than phenols and alcohols.
 - Explain why carboxylic acids are more acidic than alcohols.
 - Discuss the effect of electron-withdrawing groups on the acidity of carboxylic acids.